IB DP Design & Technology

COMPLETE CHEAT SHEET

Full Syllabus | HL + SL | Score a 7

Human Factors • Sustainability • Modelling • Materials • Innovation • UCD

20 Pages • All Topics Covered • Exam-Ready • HL Extensions Included

HOW TO USE THIS CHEAT SHEET

This cheat sheet covers the complete IB DP Design & Technology syllabus for both SL and HL students. Use it for:

- Fast topic review before class or paper practice
- Exam night revision — every key concept in one place
- Checking IB terminology and command-term responses
- HL students: purple boxes contain HL-only content

SYLLABUS MAP — COMPLETE TOPIC OVERVIEW

#	Topic	Pages	Key Content
Section 1	**Human Factors & Ergonomics**	Pages 3–4	Anthropometrics, ergonomics, inclusive design
Section 2	**Resource Management & Sustainability**	Page 5	LCA, circular economy, sustainability
Section 3	**Modelling**	Page 6	Physical, virtual, mathematical models
Section 4	**Raw Material to Final Product**	Pages 7–8	Materials, properties, manufacturing processes
Section 5	**Innovation & Design**	Page 9	Innovation types, design cycle
Section 6	**Classic Design**	Page 10	Design movements, iconic products
Section 7	**User-Centred Design**	Pages 11–12	Design thinking, user needs, evaluation
Section 8	**HL Extensions**	Pages 13–15	Advanced materials, sustainability, case studies
Exam Mastery	**Exam Strategy & Revision**	Pages 16–20	Mistakes, terms, strategy, checklists

HOW TO READ THE BOXES

- ORANGE boxes = Exam Tips (what IB actually tests)
- RED boxes = Common Mistakes (avoid these in exams)
- GREEN boxes = Quick Recall (flash-card style summaries)
- PURPLE boxes = HL Only content

SECTION 1 — HUMAN FACTORS & ERGONOMICS

CORE CONCEPTS

ANTHROPOMETRICS	ERGONOMICS
Study of human body measurements and dimensions	Design of products/systems to fit human use
Uses percentile data (5th, 50th, 95th percentile)	Focuses on comfort, efficiency, safety
Informs furniture, clothing, tool dimensions	Physical, cognitive, and environmental ergonomics
Key: design for the range, not the average	Example: keyboard layout, chair height adjustment
Static vs Dynamic anthropometrics	Reduces risk of injury and user fatigue

Percentile Data — Key Principle

- 5th Percentile: smaller users — used for minimum clearances
- 50th Percentile: average user — used as general baseline
- 95th Percentile: larger users — used for maximum reach/space
- Design for 5th–95th range to accommodate 90% of population
- NEVER design for 100% — statistically and economically impractical

PSYCHOLOGICAL FACTORS IN DESIGN

FACTOR	DESIGN IMPLICATION
Gestalt principles (proximity, similarity)	Users group related visual elements together
Affordance	Design should suggest how it is used (e.g., handle = pull)
Signifiers	Visual cues indicating possible actions (e.g., button shape)
Mental models	Users expect products to work like similar products
Cognitive load	Reduce complexity — too many choices causes decision fatigue
Colour psychology	Red = danger/stop; Green = safe/go; Blue = calm/trust

INCLUSIVE / UNIVERSAL DESIGN

- Definition: design for ALL users, regardless of age, ability, or disability
- 7 Principles of Universal Design: equitable use, flexibility, simple & intuitive, perceptible info, tolerance for error, low physical effort, size & space for approach
- Example: ramps (wheelchair + pushchair + cyclist), large-button phones
- Inclusive design vs Assistive technology — inclusive is embedded, assistive is added

EXAM TIP

- IB loves questions about anthropometric data — always state WHICH percentile and WHY
- Explain affordance with a specific product example in every answer
- Distinguish between ergonomics (fit to human) and anthropometrics (measure of human)

SECTION 1 (CONT.) — APPLICATION & EXAMPLES

REACH ZONES — WORKSTATION DESIGN

Zone	Description	Example Application
Primary zone	Forearm reach without elbow movement	Keyboard, primary controls
Secondary zone	Full arm reach	Mouse, monitor, secondary controls
Extended zone	Lean/stretch required	Rarely-used items only

GOOD VS BAD DESIGN EXAMPLES

GOOD DESIGN

- Office chair: adjustable height (5th–95th %ile)
- Pedestrian crossing: audible + visual signals
- OXO Good Grips: soft handles for arthritic users
- Smartphone: large icon option for elderly
- Car seat: adjustable lumbar + headrest

BAD DESIGN

- Fixed-height workstation ignoring user range
- Small instruction text (low legibility)
- Door with ambiguous push/pull (no signifier)
- Controls requiring both hands simultaneously
- High-contrast colour ignored for visually impaired

COMMON IB EXAM QUESTION TYPES

- Identify one ergonomic consideration for [product]
- Explain how anthropometric data was used in the design of [product]
- Evaluate the effectiveness of the design in relation to human factors
- Suggest two improvements to make [product] more ergonomic
- Outline how inclusive design principles have been applied

COMMON MISTAKE

- Confusing anthropometrics (measurement) with ergonomics (design fit)
- Saying 'designed for the average user' — IB wants a range, not average
- Forgetting to state percentile in anthropometric answers
- Listing principles without linking to a specific product example

QUICK RECALL

- Anthropometrics = measuring humans | Ergonomics = designing for humans
- 5th %ile = minimum | 95th %ile = maximum | Never design for 100%
- Affordance = what you can do | Signifier = how you know to do it
- Inclusive design: equitable, flexible, simple, low effort

SECTION 2 — RESOURCE MANAGEMENT & SUSTAINABILITY

KEY SUSTAINABILITY PRINCIPLES

- Sustainable development: meeting present needs without compromising future generations
- The 6 R's: Rethink, Refuse, Reduce, Reuse, Recycle, Repair
- Triple bottom line: People, Planet, Profit (social, environmental, economic)
- Clean production: minimise waste during manufacturing

LIFE CYCLE ANALYSIS (LCA)

1. RAW MATERIAL EXTRACTION — Mining, logging, drilling

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2. MATERIALS PROCESSING — Refining, smelting, treatment

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3. MANUFACTURING — Assembly, machining, fabrication

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4. DISTRIBUTION — Packaging, transport, storage

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5. USE PHASE — Consumer use, maintenance, repairs

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6. END OF LIFE — Disposal, recycling, landfill, incineration

LCA STAGE	ENVIRONMENTAL IMPACT EXAMPLES
Raw material extraction	Habitat destruction, water use, carbon emissions
Manufacturing	Energy consumption, toxic waste, pollution
Packaging & distribution	Plastic waste, transport emissions
Use phase	Energy consumption, consumable waste
End of life	Landfill pollution, recyclability, energy recovery

CIRCULAR ECONOMY

LINEAR ECONOMY

- Take → Make → Dispose
- Finite resource extraction
- Creates waste at end of life
- Example: single-use plastic

CIRCULAR ECONOMY

- Reduce → Reuse → Regenerate
- Resources kept in use as long as possible
- Waste becomes input for new products
- Example: Patagonia repair programme

CIRCULAR ECONOMY STRATEGY	EXAMPLE
Design for disassembly	Products designed to be easily taken apart for recycling
Industrial symbiosis	One company's waste becomes another's input
Product-as-a-service	Lease not own (e.g., Rolls Royce 'Power by the Hour')
Biomimicry	Design inspired by natural processes that produce no waste
Upcycling	Transforming waste into higher-value products

EXAM TIP

- LCA questions: always cover ALL stages, not just manufacturing
- 6 R's: IB expects you to apply them to a specific product — don't just list them
- Circular vs linear: use a real product example to contrast both
- Triple bottom line: show trade-offs — e.g., recycling costs more (profit) but saves planet

COMMON MISTAKE

- Only discussing end-of-life recycling — LCA covers ALL stages
- Confusing 'recyclable' with 'recycled' — materials may be recyclable but not actually recycled
- Ignoring the use-phase energy consumption in LCA answers

SECTION 3 — MODELLING

TYPES OF MODELS

MODEL TYPE	PURPOSE	EXAMPLES
Physical model	Test form, fit, ergonomics, aesthetics	Foam mock-up, scale model, prototype
Virtual / CAD model	Test function, movement, stress analysis	SolidWorks, AutoCAD, Fusion 360
Mathematical model	Predict performance, cost, behaviour	Stress calculations, cost formulas
Appearance model	Present aesthetics to stakeholders	Rendered images, foam models
Working prototype	Full function test before production	Beta product, pilot batch

PURPOSE AND BENEFITS OF MODELLING

BENEFIT	DETAIL
Reduces risk	Test and fail cheaply before manufacturing investment
Enables iteration	Rapid changes without costly tooling revisions
Communicates ideas	Stakeholders can visualise the product
Tests function	Stress, fit, and performance can be verified
Saves cost	Identifying issues early is far cheaper than after production
Environmental benefit	Fewer physical materials wasted in early testing stages

MODELLING IN THE DESIGN PROCESS

PROBLEM IDENTIFICATION

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DESIGN BRIEF

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RESEARCH & ANALYSIS

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IDEA GENERATION (Sketches)

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MODELLING & PROTOTYPING

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TESTING & EVALUATION

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FINAL DESIGN SOLUTION

EXAM TIP

- IB often asks: explain why a designer would use a virtual model rather than a physical one — think: cost, speed, simulation capability
- Link modelling to a stage in the design cycle — don't discuss it in isolation
- Physical vs virtual: virtual = cheaper and faster iterations; physical = tactile and ergonomic testing

QUICK RECALL

- Physical model = touch & feel | Virtual model = simulate & analyse
- Prototyping purpose: test assumptions before committing to manufacture
- All models reduce risk and cost

SECTION 4 — RAW MATERIAL TO FINAL PRODUCT

MATERIALS OVERVIEW

MATERIAL GROUP	EXAMPLES	KEY PROPERTIES
Ferrous metals	Steel, cast iron, stainless steel	Strong, magnetic, prone to corrosion (except SS)
Non-ferrous metals	Aluminium, copper, titanium	Lightweight, non-magnetic, corrosion resistant
Thermoplastics	PET, ABS, polypropylene, HDPE	Melt on heating, re-mouldable, recyclable
Thermosets	Epoxy, polyester resin, melamine	Harden permanently on curing, heat resistant
Softwoods	Pine, spruce, cedar	Fast-growing, lower cost, less dense
Hardwoods	Oak, mahogany, teak	Slow-growing, durable, high quality
Composites	Carbon fibre, GRP, plywood	Combines properties of two+ materials
Smart materials	Shape memory alloy, thermochromic	Change properties in response to stimuli
Ceramics	Porcelain, glass, alumina	Hard, brittle, heat resistant, corrosion resistant

MATERIAL PROPERTIES — KEY DEFINITIONS

PROPERTY	DEFINITION
Strength (tensile)	Ability to resist pulling forces without breaking
Hardness	Resistance to surface scratching or indentation
Toughness	Ability to absorb energy without fracturing (opposite of brittle)
Ductility	Ability to be drawn into wire without breaking
Malleability	Ability to be shaped by hammering without cracking
Elasticity	Returns to original shape after deforming force is removed
Plasticity	Permanently deforms without fracturing
Conductivity (thermal/electrical)	How easily heat or electricity passes through
Density	Mass per unit volume — key for lightweight applications
Corrosion resistance	Resistance to chemical degradation (oxidation, rust)

EXAM TIP

- Match material to application: e.g., ABS for injection moulding (cheap, tough, good surface finish)
- Composites: always explain WHICH two materials and what property each contributes
- Smart materials: know ONE example + stimulus + response for each

SECTION 4 (CONT.) — MANUFACTURING PROCESSES

FORMING PROCESSES

PROCESS	MATERIAL	KEY FEATURES
Injection moulding	Thermoplastics	High volume, accurate, fast, expensive tooling
Blow moulding	Thermoplastics	Hollow shapes (bottles), high speed
Vacuum forming	Thermoplastic sheet	Low cost tooling, thin-walled products
Die casting	Metals (zinc, aluminium)	High pressure, accurate, smooth finish
Sand casting	Metals	Low cost, large parts, rough finish
Extrusion	Metals/plastics	Continuous profile shapes (pipes, frames)
Forging	Metals	Hammered to shape — high strength grain structure
Rolling	Metals	Flat sheet/plate production
Rotational moulding	Thermoplastics	Hollow large products (tanks, toys)
Compression moulding	Thermosets/composites	Moulded under heat and pressure

PRODUCTION SCALE

PRODUCTION TYPE	CHARACTERISTICS & EXAMPLES
One-off / Bespoke	Single item, high skill, high cost, long time — e.g., custom furniture
Batch production	Groups of identical items, flexible, moderate cost — e.g., clothing
Mass production	High volume, continuous, low unit cost, specialised machinery — e.g., cars
Continuous/flow	Non-stop 24/7, e.g., steel rolling, oil refining
Just-in-time (JIT)	Stock minimised, components ordered as needed, reduces waste
Lean manufacturing	Eliminate 7 wastes: overproduction, waiting, transport, inventory, motion, defects, over-processing

QUALITY CONTROL vs QUALITY ASSURANCE

QUALITY CONTROL (QC)

- Inspects products AFTER production
- Detects defects — reactive
- Removes faulty items
- Example: final inspection, sampling

QUALITY ASSURANCE (QA)

- Manages processes DURING production
- Prevents defects — proactive
- Standards and procedures
- Example: ISO 9001, SPC monitoring

COMMON MISTAKE

- Confusing QC (after) with QA (during/before)
- Choosing a process without justifying it — always link process to material type
- Ignoring tooling costs when evaluating one-off vs mass production

QUICK RECALL

- Thermoplastics = injection moulding | Thermosets = compression moulding
- High volume + low cost = mass production | Unique + high skill = bespoke

- JIT reduces inventory costs but increases supply chain risk

SECTION 5 — INNOVATION & DESIGN

INNOVATION vs INVENTION

INVENTION

- Creating something NEW that did not exist
- May not be commercialised
- Example: first telephone (Bell, 1876)
- Often technical breakthrough

INNOVATION

- Applying ideas to create VALUE
- Practical, market-driven improvement
- Example: smartphone (2007)
- Often incremental

TYPES OF INNOVATION

TYPE	DESCRIPTION & EXAMPLE
Incremental	Small, continuous improvements — e.g., annual iPhone upgrades
Radical / Disruptive	Fundamentally changes the market — e.g., digital photography vs film
Product innovation	New or significantly improved product — e.g., electric car
Process innovation	New manufacturing or delivery method — e.g., 3D printing in medicine
Service innovation	New way of delivering value — e.g., Netflix vs video rental
Social innovation	Addresses societal challenges — e.g., low-cost prosthetics
Open innovation	Collaboration across organisations — e.g., Linux, Wikipedia

THE DESIGN CYCLE

INQUIRING & ANALYSING — Research, brief, need identification

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DEVELOPING IDEAS — Ideation, sketching, models

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CREATING THE SOLUTION — Making, construction, prototyping

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EVALUATING — Testing against criteria, feedback, iterating

- The design cycle is ITERATIVE — you return to earlier stages based on evaluation
- Design brief: a clear statement of the problem/need to be solved
- Design specification: measurable criteria the solution must meet

FACTORS INFLUENCING DESIGN

FACTOR	INFLUENCE ON DESIGN
Social	Cultural norms, user needs, accessibility
Economic	Manufacturing cost, market price, profit margin
Environmental	Sustainability, materials choice, energy use
Technological	Available processes, materials, software
Political/Legal	Safety standards, IP protection, regulations
Ethical	Labour conditions, animal testing, cultural sensitivity

EXAM TIP

- Use STEEPEL framework for design factors: Social, Technological, Economic, Environmental, Political, Ethical, Legal
- Distinguish incremental from radical innovation with named examples
- Design cycle questions: always mention that it is ITERATIVE, not linear

SECTION 6 — CLASSIC DESIGN

MAJOR DESIGN MOVEMENTS

MOVEMENT	ERA & KEY IDEA	EXAMPLES
Arts & Crafts	1880s–1910 — Handcraft, natural materials, reaction to industrialisation	William Morris textiles, hand-made furniture
Art Nouveau	1890–1910 — Organic curves, nature-inspired, decorative	Tiffany lamps, Gaudi architecture
Bauhaus	1919–1933 — Form follows function, minimal ornament, industrial	Marcel Breuer chairs, Mies furniture
Streamlining	1930s–40s — Aerodynamic form, speed, modernity	Hoover 150 vacuum, Chrysler Airflow car
Modernism	1950s–70s — Clean lines, functionalism, universal design	Charles Eames chairs, Dieter Rams products
Postmodernism	1970s–90s — Humour, historical references, eclecticism	Memphis Group, Philippe Starck's juicers
Minimalism	1990s–present — Simplicity, whitespace, restraint	Muji products, early Apple iMac

ICONIC PRODUCTS & DESIGNERS

DESIGNER / PRODUCT	SIGNIFICANCE
Dieter Rams — Braun products	10 principles of good design; influenced Apple's Jony Ive
Charles & Ray Eames — Lounge Chair	Plywood moulding; form AND function; still in production
Jonathan Ive — Apple iMac/iPod/iPhone	Minimalist aesthetic, user-centred, seamless integration
Philippe Starck — Juicy Salif juicer	Postmodern: iconic but controversial; form over function
Buckminster Fuller — Geodesic dome	Maximum structural efficiency with minimum material
Ettore Sottsass — Memphis Group	Playful, colourful postmodern design movement

DIETER RAMS — 10 PRINCIPLES OF GOOD DESIGN

- Innovative
- Makes a product useful
- Aesthetic
- Makes a product understandable
- Unobtrusive
- Honest
- Long-lasting
- Thorough down to the last detail
- Environmentally friendly
- As little design as possible

EXAM TIP

- For classic design questions: name movement + era + key aesthetic + named example
- Dieter Rams principles often appear in IB evaluation questions
- Compare two movements: Bauhaus (function) vs Postmodernism (form/decoration)

QUICK RECALL

- Bauhaus: Form follows function | Postmodern: decoration returns
- Eames: ergonomics + aesthetic | Rams: honest + minimal
- Modernism = universal, timeless | Postmodernism = personal, playful

SECTION 7 — USER-CENTRED DESIGN (UCD)

WHAT IS UCD?

- User-centred design places USER needs at the centre of every design decision
- Design is iterative: test → feedback → revise → repeat
- Contrast: technology-driven (what we CAN make) vs user-driven (what users NEED)

STAKEHOLDER ANALYSIS

STAKEHOLDER TYPE	ROLE & INFLUENCE
Primary users	People who directly use the product — highest design priority
Secondary users	Indirectly affected (e.g., maintenance staff, bystanders)
Clients/Clients	Who commissions the design — may differ from user
Manufacturers	Must be able to produce the design economically
Retailers	Packaging, display, and sales requirements
Society/Environment	Broader impact — sustainability, cultural sensitivity

USER RESEARCH METHODS

METHOD	PURPOSE & EXAMPLE
Interviews	Deep qualitative insight; explore motivations and needs
Surveys/Questionnaires	Quantitative data; large sample; e.g., 5-point Likert scale
Observation	See real use context; uncover hidden behaviours
Focus groups	Group discussion; generate ideas; reveal shared needs
User personas	Fictional representative users to guide design decisions
Empathy mapping	Understand what users think, feel, say, do
Journey mapping	Map user experience across all touchpoints

SECTION 7 (CONT.) — DESIGN THINKING & EVALUATION

THE DESIGN THINKING PROCESS

1. EMPATHISE — Understand the user; research, observe, listen

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2. DEFINE — Identify the core problem (problem statement / HMW)

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3. IDEATE — Generate diverse ideas without judgement (brainstorm)

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4. PROTOTYPE — Build quick, low-fidelity representations to test

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5. TEST — Try with real users; gather feedback; iterate

EVALUATION CRITERIA

CRITERION	EVALUATION APPROACH
Function	Does it perform its intended task reliably?
Ergonomics	Is it comfortable, safe, and efficient to use?
Aesthetics	Is it visually appealing to the target user?
Sustainability	What is the environmental impact across its lifecycle?
Cost	Is it affordable for the target market?
Safety	Does it meet relevant safety standards and regulations?
Durability	Will it last under expected use conditions?
User feedback	What do real users say after testing?

TESTING METHODS

TEST TYPE	WHEN & WHY USED
Usability testing	Observe users completing tasks with prototype — reveal pain points
A/B testing	Compare two versions to determine which performs better
Heuristic evaluation	Expert review against usability principles (e.g., Nielsen's heuristics)
Physical testing	Mechanical, stress, durability testing of product
User acceptance testing	Does product meet user requirements before launch?

EXAM TIP

- UCD questions: always link design decision back to USER needs or feedback
- Evaluation: use specific criteria, not vague statements — 'the handle should fit 95th %ile grip width'
- Design thinking is not linear — emphasise iteration in exam answers

COMMON MISTAKE

- Confusing the client (who pays) with the user (who uses) — they are often different
- Evaluating only aesthetics — IB wants functional, ergonomic, sustainability criteria

– Describing design thinking as a one-way process — it is iterative

SECTION 8 — HL EXTENSIONS

HL ONLY

- All content in this section is HL ONLY — SL students may skip

ADVANCED & SMART MATERIALS (HL)

MATERIAL	STIMULUS / RESPONSE	APPLICATION
Shape memory alloy (SMA)	Temperature → returns to original shape	Stents, spectacle frames, actuators
Thermochromic	Temperature → colour change	Thermometers, mood rings, smart labels
Photochromic	Light → colour/opacity change	Transition lenses, smart glass
Piezoelectric	Mechanical stress → electric current (& vice versa)	Sensors, actuators, energy harvesting
Magnetorheological (MR)	Magnetic field → fluid viscosity change	Smart car dampers, prosthetics
Quantum dots	Size-tuned semiconductor nanocrystals → tunable colour/light emission	QLED displays, solar cells
Graphene	Single atom layer carbon — extreme strength, conductivity	Electronics, composites, filtration
Aerogel	Ultra-low density silica — extreme thermal insulation	Mars rovers, thermal glazing

NANOTECHNOLOGY IN DESIGN (HL)

- Nanotechnology = engineering at 1–100 nanometre scale
- Applications: nano-coatings (scratch resist), nano-textiles (waterproof), nano-medicine
- Benefit: properties emerge at nano-scale that don't exist at macro-scale
- Concern: health risks of nanoparticles not fully understood; regulation lags behind innovation

SUSTAINABLE DESIGN — DEEPER CONCEPTS (HL)

CONCEPT	DETAIL
Biomimicry	Design inspired by nature (Velcro = burdock; Shinkansen nose = kingfisher)
Cradle-to-Cradle (C2C)	All materials are either biological or technical nutrients — zero waste
Industrial ecology	Model industrial processes on natural ecosystems; waste = resource
Factor 4 / Factor 10	Quadruple/10x efficiency of resource use while maintaining output
Ecodesign strategies	Reduce material, extend life, ease of disassembly, less toxic
Carbon footprint	Total GHG emissions over product lifecycle; measured in CO2e
Water footprint	Total freshwater used in product lifecycle (embedded water)

SECTION 8 (CONT.) — HL INNOVATION CASE STUDIES

CASE STUDY 1: TESLA — DISRUPTIVE INNOVATION

- Product: Tesla Model S (2012) — first premium electric car with 400+ km range
- Innovation type: Radical/disruptive — challenged ICE automotive market
- Materials: High-strength aluminium body, lithium-ion battery pack
- Process innovation: Software over-the-air updates; direct sales model (no dealerships)
- Sustainability: Zero tailpipe emissions; however mining of lithium raises LCA concerns
- UCD: Large touchscreen replacing buttons — controversial ergonomics debate

CASE STUDY 2: PATAGONIA — CIRCULAR ECONOMY LEADER

- Philosophy: 'Don't buy this jacket' campaign — anti-consumption messaging
- Worn Wear programme: repair and resell used products
- Materials: Recycled polyester from plastic bottles; organic cotton
- B Corp certification: meets social and environmental performance standards
- Business model innovation: sell fewer but higher-quality, longer-lasting products

CASE STUDY 3: OXFAM/AIDPOD — DESIGN FOR DEVELOPMENT

- Context: Designing for low-income / developing world markets
- Constraints: Low cost, durable, repairable locally, no electricity required
- Design principle: Appropriate technology — match solution to local resources
- Example: Life Straw — portable water filtration for disaster zones
- Impact assessment: social benefit, not just commercial profitability

HL ONLY

- HL exam often asks to evaluate a named product against sustainability/innovation criteria
- Case study questions require specific product names, dates, and measurable outcomes
- For biomimicry: name the natural organism + feature + how it was applied

EXAM MASTERY — COMMON MISTAKES TO AVOID

TOP 15 COMMON IB DESIGN & TECHNOLOGY MISTAKES

1	**Using vague words**	Say 'reduces carbon emissions by eliminating combustion' not 'better for environment'
2	**Confusing anthropometrics and ergonomics**	Anthropometrics = data collection; Ergonomics = applying data to design
3	**Designing for the average user**	IB always wants a RANGE (5th–95th percentile), not the 50th
4	**Listing 6 R's without application**	Apply each R to a specific product with a real consequence
5	**Treating LCA as just recycling**	LCA covers raw extraction, manufacturing, distribution, use, AND end-of-life
6	**Treating design cycle as linear**	Always state it is ITERATIVE — you return to earlier stages
7	**QC vs QA confusion**	QC = check after | QA = prevent before/during
8	**Choosing material with no justification**	Always link property to function: 'aluminium is chosen because it is lightweight yet strong'
9	**Circular vs linear economy confusion**	Linear = take-make-dispose | Circular = keep in use, regenerate
10	**Client vs user confusion**	Client commissions the design; user actually uses it — they may be different people
11	**Describing not evaluating**	Evaluation requires judgement: 'this IS effective because...'; not just 'this design has a handle'
12	**Ignoring command terms**	Evaluate = pros/cons + judgement | Explain = reason given | Identify = name only
13	**No specific examples in classic design**	Always name designer + product + era — never generic ('Bauhaus was minimalist')
14	**Smart materials without stimulus/response**	Always state: material + stimulus + response — e.g., SMA + heat → shape recovery
15	**Not quantifying sustainability claims**	Use metrics: '35% recycled content', 'carbon neutral by 2030', not 'it is sustainable'

EXAM STRATEGY — HOW TO STRUCTURE ANSWERS

IB COMMAND TERMS — WHAT THEY REALLY MEAN

COMMAND TERM	WHAT YOU MUST DO + EXAMPLE STRUCTURE
Identify	Name only — 1 word or phrase is sufficient. No explanation needed.
State	Give a specific fact or value — brief, factual. Example: 'The material is ABS.'
Outline	Give main points briefly — no deep explanation needed.
Describe	Give detailed characteristics — what it looks like or does. Use adjectives.
Explain	Give reason(s) — WHY or HOW. Use 'because', 'therefore', 'this means'.
Discuss	Give arguments for AND against, then a balanced conclusion.
Evaluate	Assess pros and cons with evidence; make a justified judgement.
Analyse	Break into parts and examine each; show relationships and implications.
Justify	Give valid reasons or evidence to support a conclusion.
Compare	Show BOTH similarities AND differences — use a structured format.
Contrast	Show ONLY the differences between two things.
Suggest	Propose a solution or improvement — any reasonable answer accepted.

STRUCTURING EXTENDED ANSWERS

POINT — Make a clear, specific statement

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EVIDENCE — Name a material, product, process, or data

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EXPLANATION — Explain WHY this is relevant (because/therefore)

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LINK — Connect back to the question focus

- For EVALUATE questions: Structure as Advantage → Evidence → Disadvantage → Evidence → Conclusion
- For DISCUSS questions: Stakeholder A view → Stakeholder B view → Balanced judgement
- Mark allocation guide: 1 mark = 1 specific point; 2 marks = point + explanation

EXAM TIP

- For every claim you make, ask: 'So what? Why does this matter?' — then write that too
- Use subject-specific vocabulary in EVERY answer — IB marks reward precise terminology
- Read the mark scheme language: 'award 1 mark for...' tells you exactly what examiners want

KEY TERMS GLOSSARY

TERM	DEFINITION	CONTEXT
Anthropometrics	Science of measuring human body dimensions	Ergonomics, HF
Ergonomics	Designing products/environments to fit human use	HF
Inclusive design	Design accessible to ALL users regardless of ability	HF, UCD
Affordance	Property of an object that suggests its use	UCD, HF
Signifier	Perceptible signal showing possible actions	UCD
LCA	Life cycle analysis — evaluating environmental impact from cradle to grave	Sustainability
Circular economy	Economy that keeps resources in use and regenerates natural systems	Sustainability
6 R's	Rethink, Refuse, Reduce, Reuse, Recycle, Repair	Sustainability
Carbon footprint	Total GHG emissions attributable to a product or activity	Sustainability
Biomimicry	Innovation inspired by biological strategies	Sustainability, Innovation
Prototype	Early model built to test function, form, or process	Modelling
Thermoplastic	Plastic that softens on heating, re-mouldable	Materials
Thermoset	Plastic that cures permanently; cannot be re-melted	Materials
Composite	Material made from two+ constituent materials	Materials
Smart material	Material that responds to environmental stimuli	Materials, HL
Injection moulding	Plastic forced under pressure into a mould cavity	Manufacturing
JIT	Just-in-time: materials ordered only as needed	Manufacturing
QC vs QA	Quality Control (check after) vs Quality Assurance (prevent before)	Manufacturing
Radical innovation	Fundamentally new product/process disrupting existing market	Innovation
Incremental innovation	Small, continuous improvement to existing product	Innovation
UCD	User-centred design: design driven by user needs and feedback	UCD
Stakeholder	Any person or group affected by the design outcome	UCD
Design thinking	Empathise → Define → Ideate → Prototype → Test	UCD
Bauhaus	1919 design movement: form follows function, industrial aesthetic	Classic Design
SMA	Shape memory alloy: recovers shape on heating	HL Materials
Cradle-to-Cradle	Design system where all materials cycle as nutrients, zero waste	HL Sustainability

QUICK REVISION — ALL TOPICS AT A GLANCE

SECTION 1 — HUMAN FACTORS

- Anthropometrics: measure humans | Ergonomics: design for humans
- Percentiles: 5th (small), 50th (average), 95th (large) — design for range
- Affordance: suggests use | Signifier: shows HOW to use
- Inclusive design: 7 principles — equitable, flexible, simple, perceptible, tolerant, low effort, size/space

SECTION 2 — SUSTAINABILITY

- 6 R's: Rethink, Refuse, Reduce, Reuse, Recycle, Repair
- LCA: extraction → processing → manufacturing → distribution → use → end-of-life
- Circular economy: closed-loop, no waste | Linear: take-make-dispose
- Triple bottom line: People, Planet, Profit

SECTION 3 — MODELLING

- Physical model = tactile test | Virtual/CAD = simulate, iterate fast
- Modelling purpose: reduce risk, communicate, iterate cheaply

SECTION 4 — MATERIALS & MANUFACTURING

- Thermoplastics: melt → mould → re-melt | Thermosets: cure permanently
- Composites = two materials → combined properties
- Production: one-off → batch → mass → continuous
- QC = check after | QA = prevent before

SECTION 5 — INNOVATION

- Invention = new | Innovation = value from ideas
- Design cycle: Inquire → Develop → Create → Evaluate (ITERATIVE)

SECTION 6 — CLASSIC DESIGN

- Bauhaus: form follows function | Postmodern: decoration & humour
- Dieter Rams 10 principles — minimalist, honest, long-lasting

SECTION 7 — UCD

- Design thinking: Empathise → Define → Ideate → Prototype → Test
- Evaluation criteria: function, ergonomics, aesthetics, sustainability, cost, safety

FINAL CHECKLIST — NIGHT BEFORE THE EXAM

KNOWLEDGE CHECKLIST — CAN YOU...

- Define anthropometrics AND ergonomics with a product example?
- Explain the 5th/50th/95th percentile and why we never design for 100%?
- List all 6 stages of a Life Cycle Analysis?
- Explain the difference between circular and linear economy?
- Name 3 types of models and explain when each is used?
- List 5 material properties with definitions and examples?
- Name 4 manufacturing processes with material types?
- Distinguish QC from QA with a manufacturing example?
- Explain incremental vs radical innovation with named products?
- Name 3 design movements with era, aesthetic, and key designer?
- Describe the 5 stages of design thinking?
- List 4 evaluation criteria for a product?
- (HL) Name 4 smart materials with stimulus and response?
- (HL) Explain Cradle-to-Cradle and give an example?

EXAM DAY STRATEGY

- Read the command term FIRST — it tells you exactly what to do
- For every mark, write one specific point with evidence
- Use precise IB terminology in every answer
- Draw and annotate diagrams where possible — labelled diagrams earn marks
- For evaluate/discuss: always include a conclusion with a justified judgement
- Time management: 1 mark = ~1 minute; 6-mark question = 6 minutes
- Check: have you answered every part of a multi-part question?
- Use STEEPEL for design factors: Social, Tech, Economic, Environmental, Political, Ethical, Legal

YOU'VE GOT THIS.

Use precise terminology. Apply to real examples. Structure your answers. Score a 7.